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SUPPOSE WE HAD a measure of "health status" for every individual in the world. Then we could calculate an inequality index (such as a Gini coefficient) from those data to obtain an estimate of global health inequality. Alternatively, we could estimate global health inequality in two steps, first calculating the level of health inequality among individuals within each country, then the level of inequality in average health status across or between countries. Using an additively decomposable inequality index, global inequality (I_G) is the sum of inequality between countries (I_B) and the average level of inequality within countries (I_W):

$$I_G = I_B + I_W . \tag{1}$$

Equation 1 is similar to analysis of variance, where total variance in some outcome variable is divided into the sum of population-weighted between-group and within-group components. Here total inequality (a type of relative variance) is divided into the sum of population-weighted between-group and within-group components, with countries as groups. Recent research on global health inequality has centered on the second component, health inequality within countries (e.g., Gakidou and King 2002; Gakidou, Murray, and Frenk 2000; Murray, Gakidou, and Frenk 1999; WHO 2000). This note is about the first component, health inequality between countries.

Other studies of cross-country disparities or inequalities in health (e.g., Anand and Ravallion 1993; Pritchett and Summers 1996; Preston 1975; Subramanian, Belli, and Kawachi 2002; Wilkinson 1996; Wimberley 1990) have focused on why population health is better in some countries than in others. Unlike these studies, our main objective is to provide a thorough empirical description of the recent trend in between-country health inequality. Has health inequality across countries risen or declined over the past 20 years? How does the recent trend compare to historical trends? Which countries or regions have contributed most to the trend? Answers to such questions are critical to our understanding of between-country and global health inequality at the dawn of the twenty-first century.

Basic concepts and terms

By *between-country health inequality* we mean the uneven distribution of health across countries. We focus on one type of between-country health inequality: the uneven distribution of life expectancy across countries. We use life expectancy as a measure of population health—first, because it is one of few indicators available for a near-universe of the world's population and, second, because it is a more intuitively meaningful measure of population health than some other possible measures, such as anthropomorphic data on human body mass or stature (e.g., Fogel 1993; Steckel 1995, 2001; Steckel and Floud 1997). Estimates of life expectancy would ideally be adjusted for years of healthy life lost due to disability (Crimmins, Saito, and Ingegneri 1989, 1997; Mathers et al. 2001; Murray and Lopez 1996, 1997), but here we use just total life expectancy because it provides the broadest possible coverage across countries and over time.

Health researchers sometimes distinguish between two approaches to the measurement of health inequality (Kawachi, Subramanian, and Almeida-Filho 2002; Wolfson and Rowe 2001). One approach defines health inequality as differences in average health status across population subgroups. Studies that examine health disparities by age, sex, race, or social class are common examples of this approach. Other examples are studies that analyze health disparities between rich and poor countries, such as Preston's (1975) classic study of the cross-national relationship between average national income and life expectancy. The second approach defines health inequality as the uneven distribution of health across all units in a population, independent of population subgroup (e.g, Gakidou et al. 2000; Gakidou and King 2002; Murray et al. 1999; Pradhan et al. 2003; Shkolnikov, Andreev, and Begun 2003; Wilmoth and Horiuchi 1999). In this approach, inequality is viewed as a type of overall relative variance and measured with inequality indexes such as the Gini coefficient. Wolfson and Rowe (2001) dub this the "univariate" approach to measuring health inequality. It is akin to the methods used in the large social scientific literature on inequality in aggregate income distributions (e.g., Allison 1978; Cowell 1995; Jenkins 1991; Sen 1997). In this study we use the second approach, examining the uneven distribution of life expectancy across countries without reference to other national characteristics.

Cross-country evidence

In the early nineteenth century, the range in life expectancy across countries was small. The best historical estimates suggest that life expectancy at birth ranged from the low 20s in the least-healthy countries to the low 40s in the healthiest countries (Maddison 2001). Inequality across countries then increased through the second half of the nineteenth and the early twentieth century, as a "mortality revolution" (Easterlin 1996) swept through the West. In the early 1800s, average life expectancy in Western Europe was perhaps 36 years; by 1900 it had reached 46 years, and by 1950 nearly 70 years (Maddison 2001). A similar trend occurred in the Western offshoots—Australia, New Zealand, Canada, and the United States.

Inequality appears to have peaked in the twentieth century between the two world wars (Easterlin 1996: 70–72). At that peak, life expectancy was perhaps twice as great in the West as in sub-Saharan Africa, South Asia, and China. East Asia (outside China), Latin America, and Eastern Europe lay between the two extremes. The second half of the twentieth century then saw a sharp drop in inequality across countries, as the mortality decline spread to other parts of the world. In this period, the largest gains in life expectancy occurred in regions where levels of life expectancy were lowest. Life expectancy has increased by 36 percent in Africa, by over 34 percent in Latin America and the Caribbean, and by almost 60 percent in Asia since the mid-twentieth century (UNPD 2002). To be sure, there are still large differences in life expectancy from country to country, but these differences are on average smaller today than they were 50 or 100 years ago.[1]

Most analysts assume that once life expectancies start to converge across countries, the trend does not reverse. Empirical evidence about the recent inequality trend, however, is mixed. Wilson (2001) finds that life expectancies have converged rapidly across countries since 1950, the year his analysis begins. The result, he argues, is a world where national boundaries are of "diminishing demographic relevance" (p. 168). Mayer (2001) also reports a trend of convergence, but only within—not between—groups of rich and poor countries. The world's poorest countries are converging around a very low level of life expectancy (perhaps 45 or 50 years), while the richer countries are converging around much higher levels (perhaps 75 or 80 years). If that is the case, the long-term decline in between-country inequality may have halted or even reversed in the late twentieth century. The effects of convergence within groups of countries may have been overshadowed by the effects of the persistent divide between groups.

Estimates of average life expectancy (both sexes combined) in 1980, 1990, and 2000 are shown in Table 1, for the world as a whole and for eight major regions and two countries.[2] Countries are weighted by population size in all these estimates. The two right-hand columns of the table

TABLE 1 World and regional trends in life expectancy at birth (both sexes combined), 1980–2000

	Life expectancy			Percent change	
Region	1980	1990	2000	1980–1990	1990–2000
Western Europe	73.9	76.1	78.0	+3.1	+2.4
Transition economies	68.1	69.3	68.0	+1.8	−1.9
Western offshoots	73.8	75.5	77.4	+2.3	+2.4
Latin America and the Caribbean	64.7	68.0	70.4	+5.1	+3.6
Middle East and North Africa	59.2	65.0	68.5	+9.7	+5.4
Sub-Saharan Africa	47.6	50.0	46.5	+5.0	−7.0
South Asia	53.4	58.3	62.1	+9.1	+6.5
East Asia (excluding China and Japan)	59.7	65.1	68.5	+8.9	+5.2
Japan	76.1	78.8	80.7	+3.6	+2.4
China	66.8	68.9	70.3	+3.0	+2.0
World	62.5	65.2	66.4	+4.3	+1.9

Western Europe (19 countries): Austria, Belgium, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Malta, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom.
Transition economies (28 countries): Albania, Armenia, Azerbaijan, Belarus, Bulgaria, Croatia, Czech Republic, Estonia, Georgia, Hungary, Kazakhstan, North Korea, Kyrgyz Republic, Laos, Latvia, Lithuania, Moldova, Mongolia, Poland, Romania, Russian Federation, Slovak Republic, Slovenia, Tajikistan, Turkmenistan, Ukraine, Uzbekistan, Yugoslavia.
Western offshoots (4 countries): Australia, Canada, New Zealand, United States.
Latin American and the Caribbean (29 countries): Argentina, Bahamas, Barbados, Belize, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Guyana, Haiti, Jamaica, Mexico, Netherlands Antilles, Nicaragua, Panama, Paraguay, Peru, Puerto Rico, St. Lucia, Suriname, Trinidad and Tobago, Uruguay, Venezuela.
Middle East and North Africa (20 countries): Algeria, Bahrain, Brunei, Cyprus, Egypt, Iran, Iraq, Israel, Kuwait, Lebanon, Libya, Morocco, Oman, Qatar, Saudi Arabia, Syria, Tunisia, Turkey, United Arab Emirates, Yemen.
Sub-Saharan Africa (46 countries): Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Central African Republic, Chad, Comoros, Democratic Republic of the Congo, Congo Republic, Côte d'Ivoire, Djibouti, Equatorial Guinea, Eritrea, Ethiopia, Gabon, The Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, Senegal, Sierra Leone, Somalia, South Africa, Sudan, Swaziland, Tanzania, Togo, Uganda, Zambia, Zimbabwe.
South Asia (8 countries): Afghanistan, Bangladesh, Cambodia, India, Myanmar, Nepal, Pakistan, Sri Lanka.
East Asia (13 countries): Fiji, Hong Kong, Indonesia, South Korea, Malaysia, New Caledonia, Papua New Guinea, Philippines, Samoa, Singapore, Solomon Islands, Thailand, Vietnam.
SOURCE: World Bank (2002).

show the percent change in life expectancy from 1980 to 1990 and from 1990 to 2000.

Three trends observable in Table 1 bear on the subject of this note. First, the long-term trend of rising global life expectancy continued through the late twentieth century. Global life expectancy increased by about 4.3 percent from 1980 to 1990 and by 1.9 percent from 1990 to 2000. The largest recent gains occurred in the Middle East/North Africa, South Asia, and East Asia, three regions where life expectancy was below the world average in 1980. Gains were smaller in regions such as Western Europe, the Western offshoots, and Japan, where life expectancy was already very high.

Second, in the period 1990–2000 life expectancy declined in sub-Saharan Africa (by 7 percent) and the transition economies (by 1.9 percent). In sub-Saharan Africa the decline is mainly due to the HIV/AIDS epidemic

(Buvé, Bishikwabo-Nsarhaza, and Mutangadura 2002; UNAIDS 1998, 2002). In the transition economies the reasons are harder to pin down (Marmot and Bobak 2000), but are presumably connected to the collapse of the Communist regimes and the breakup of the former Soviet Union.

Third, because life expectancy changed at different rates across regions, the shape of the international distribution of life expectancy changed too. In 1980, life expectancy was above the world average in six of the ten regions listed in Table 1, and below average in the other four. The gap in life expectancy between the top and bottom of the distribution (Japan and sub-Saharan Africa) was about 29 years. By 2000, however, sub-Saharan Africa and South Asia were the only two regions below the world average, and the gap between the top and bottom had increased to 34 years. In short, the distribution of life expectancy across regions had become wider and more skewed.

Overall, the trends of faster-than-world-average growth in life expectancy in the Middle East/North Africa, South Asia, and East Asia no doubt reduced the level of inequality across countries by compressing the distance between life expectancies in these regions and the global average. Yet the declining life expectancy in sub-Saharan Africa after 1990 boosted inequality by stretching the bottom tail of the distribution. To determine which of these effects dominated, we need to analyze the recent between-country inequality trend more formally.

Measuring inequality across countries

We use four standard measures of inequality, each with different properties: the Gini coefficient, the Theil index, the mean logarithmic deviation (MLD), and the squared coefficient of variation (CV^2). Although these indexes generally are used to measure inequality in income distributions, they can also be used in the context of other ratio-level variables, including average life expectancy.[3] In the following equations, p_j is the jth country's share of the world's total population (hence $\Sigma_j p_j = 1.0$), and r_j is life expectancy in country j divided by average global life expectancy (i.e., $r_j = X_j / \Sigma_j p_j X_j$, where X_j is life expectancy in country j). The term r_j is the *life-expectancy ratio*.

On the basis of this notation, we express the four inequality indexes as follows (Firebaugh 2003: 82–83):

$$\begin{aligned} \text{Gini coefficient} &= \Sigma_j p_j r_j (q_j - Q_j) \\ \text{Theil index} &= \Sigma_j p_j r_j \ln(r_j) \\ \text{MLD} &= \Sigma_j p_j \ln(1/r_j) \\ \text{CV}^2 &= \Sigma_j p_j (r_j - 1)^2 , \end{aligned} \qquad (2)$$

where q_j is the proportion of the world's population in countries where life expectancy is lower than in country j, and Q_j is the proportion of the world's population in countries where life expectancy is greater than in country j.

All four indexes are calibrated to zero when life expectancy is distributed evenly across countries—that is, when life expectancy in each country is equal to the world average. The index values then increase as the level of inequality across countries increases. The indexes differ in their sensitivity to changes in different parts of the distribution. Compared with the other indexes, the Gini coefficient is relatively sensitive to change in the middle of the distribution; both the Theil index and CV^2 are relatively sensitive to change at the top of the distribution; and the MLD is relatively sensitive to change at the bottom of the distribution.[4] Because of this, each index yields a somewhat different estimate of change in between-country inequality, depending on how the distribution of life expectancy has changed.

The four equations above reduce to a single expression (Firebaugh 1999):

$$Inequality = \Sigma_j p_j f(r_j), \tag{3}$$

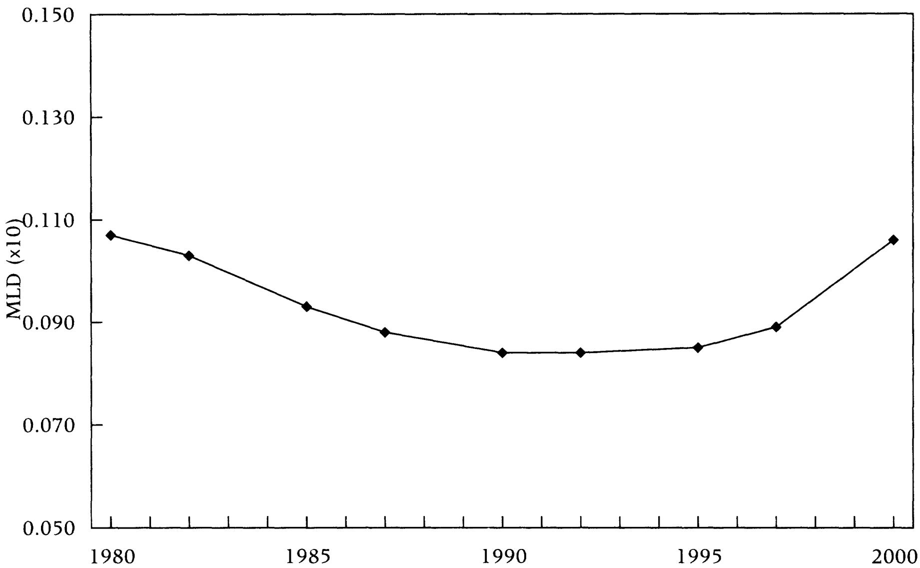
where the terms j, p_j, and r_j are the same as above, and f is the functional form used to transform the life-expectancy ratios. Equation 3 shows that between-country inequality as measured by any common inequality index is determined by just two terms, population shares (p_j's) and life-expectancy ratios (r_j's). The population shares simply weight countries by the relative size of their national populations,[5] so the life-expectancy ratios are ultimately the key term for measuring inequality. Each index first transforms the ratio with a unique specification of the mathematical function, $f(r_j)$, then averages across the transformed values. The transformations for each index converge to zero as the life-expectancy ratio approaches 1.0. Put another way, for each index, $f(r_j)$ increases in absolute value as r_j moves away from 1.0. The greater the distance of the average ratio from 1.0, the greater the level of inequality across countries.

Between-country health inequality, 1980–2000

Figure 1 depicts the trend in between-country health inequality from 1980 to 2000 as measured by the MLD. Inequality for each year is calculated from estimates of life expectancy (both sexes combined) and population size for a constant panel of 169 countries (see Table 1 for a list), which together account for a near-universe of the world's population. Data are from the World Bank (2002). Results of the analysis confirm the distributional trends implied by Table 1: between-country inequality declined from 1980 to about 1992, but then increased from 1992 to 2000 in a significant reversal of the long-term trend. According to these estimates, the level of between-country health inequality was about the same in 2000 as it was in 1980.

Table 2 shows that the trend is robust across the four inequality indexes. All four indicate a turnaround in the inequality trend beginning in

FIGURE 1 Trend in between-country health inequality, 1980–2000



NOTES: MLD (mean logarithmic deviation) is an index of inequality (see text). Trend is based on life expectancy and population data for 169 countries from World Bank (2002); see Table 1 for list of countries.

the early 1990s. The table also suggests where to look for an explanation of the upturn. The largest gain is registered by the MLD, which is more sensitive than the other indexes to change at the bottom of the distribution.

TABLE 2 Trend in between-country health inequality (both sexes), 1980–2000

	Index (x10)			
Year	Gini coefficient	Theil index	MLD	CV^2
1980	.797	.103	.107	.200
1982	.783	.099	.103	.193
1985	.740	.089	.093	.174
1987	.716	.084	.088	.163
1990	.691	.080	.084	.155
1992	.680	.079	.084	.153
1995	.680	.080	.085	.153
1997	.688	.084	.089	.160
2000	.730	.099	.106	.185
Percent change, 1980–90	–13.3	–22.3	–21.5	–22.5
Percent change, 1990–2000	+5.6	+23.8	+26.2	+19.4

NOTES: See text for description of inequality measures. Estimates are based on life expectancy and population data for 169 countries from World Bank (2002); see Table 1 for list of countries.

Decomposition analysis

We use decomposition and standardization techniques to answer two key questions. First, did between-country health inequality rise in the 1990s primarily because life expectancy changed at different rates across countries or because populations grew faster in countries with unusually low (or unusually high) life expectancies? Second, was the trend dominated by a few influential regions or countries?

Recall (from equation 3, above) that change in inequality is a function of change in population shares and in life-expectancy ratios. More formally, change in an inequality index between any two points, t and $t+1$, can be divided into two additive components, one reflecting the effect of changing life-expectancy ratios, the other reflecting the effect of changing population shares. The former is called a growth effect, the latter an allocation effect.

The decomposition formula for the MLD, chosen because it decomposes more readily than the other indexes, is as follows:

$$\Delta MLD = \Sigma_j\left(\overline{r_j} - \overline{\ln r_j}\right)\Delta p_j + \Sigma_j\left(\overline{p_j r_j} - \overline{p_j}\right)\Delta \ln X_j, \quad (4)$$

where the terms j, X_j, p_j, and r_j are the same as above, Δ is the difference operator (i.e., $\Delta MLD = MLD(t+1) - MLD(t)$), and the over-bar indicates an average of the variable across the two time-points (e.g., $\overline{r_j} = [r_j(t+1) + r_j(t)]/2$). Equation 4 is derived from the formulas given in Mookherjee and Shorrocks (1982) and Jenkins (1995).[6] The first term on the right-hand side of the equation is the allocation effect, the second term is the growth effect.

To understand how between-country inequality can change as a result of population growth independent of change in life-expectancy ratios, suppose life expectancy grows at the same rate for all countries—so life-expectancy ratios are constant—but population grows fastest in countries where life expectancy is close to the world average. Then the middle of the distribution "fattens" relative to the tails of the distribution, and inequality declines. By the same logic, inequality increases if populations grow fastest in countries in the tails of the distribution (countries that stand out with either high or low life expectancies relative to the world average). The allocation effect refers to change in between-country inequality due specifically to the changing shape of the population-weighted life-expectancy distribution across countries as populations grow faster in some countries than in others.

From 1990 to 2000, between-country health inequality grew by 26.2 percent as measured by the MLD (from Table 2). Using the decomposition formula (equation 4), we find that a growth effect accounts for about 75 percent of this increase, an allocation effect for only about 25 percent. That is, most of the increase in between-country health inequality occurred because life expectancy grew at different rates across countries. Cross-country differences in rates of population growth had a much smaller effect on the population-weighted inequality trend.

Given the nature of the recent relationship between population growth and life expectancy, the small size of the allocation effect is not surprising. In the 1990s, there was a strong negative relationship between life expectancy and national population growth: for the 169 countries in this analysis, the bivariate relationship between 1990 life expectancy and 1990–2000 population growth yields a correlation coefficient of –0.61. That is, population has tended to grow fastest in countries where life expectancy is low and slowest in countries where life expectancy is high. As a result, population growth in the middle of the international distribution of life expectancy has not differed greatly from the *average* growth in the two tails, leaving little net effect of differential population growth on the overall inequality trend.

Growth and allocation effects, by region

The decomposition method cannot be used to further divide the overall growth and allocation effects into effects for specific countries or regions (Goesling 2003). One way to do so is by comparing the actual change in between-country inequality with the counterfactual case in which the rate of population growth or of life expectancy growth for a region is set at the world average (Berry, Bourguignon, and Morrisson 1983; Bourguignon and Morrisson 2002; Sala-i-Martin 2002). The difference between the actual change in between-country inequality and the "predicted" change under the counterfactual condition is taken as an estimate of the growth or allocation effect for the region. Repeating the exercise for each region in turn, we can determine which regions contributed most to the overall trend and indicate the direction of each region's contribution.

Table 3 reports the results for growth effects and Table 4 for allocation effects. The bottom row of each table lists the observed change in between-country health inequality from 1990 to 2000, as measured by each of the four inequality indexes (from Table 2). The other rows show the predicted change in between-country inequality under the counterfactual (world-average) conditions. The top row of Table 3, for instance, shows the predicted change in between-country health inequality if life expectancy in Western Europe had grown by 1.9 percent from 1990 to 2000—the world-average rate—instead of 2.4 percent—its actual rate (from Table 1). Similarly, each row in Table 4 reports the predicted change in between-country health inequality after setting population growth for the corresponding region at the world-average rate.

TABLE 3 Growth effects on health inequality, by region, 1990–2000: Predicted change in indexes of inequality with rate of growth in life expectancy set at the world average (percent)

	Index			
Region	Gini coefficient	Theil index	MLD	CV^2
Western Europe	+5.2	+21.7	+25.5	+18.4
Transition economies	+4.7	+22.8	+26.9	+19.3
Western offshoots	+5.3	+21.8	+25.6	+18.6
Latin America and the Caribbean	+5.7	+21.9	+25.5	+18.8
Middle East and North Africa	+6.3	+22.5	+26.1	+19.5
Sub-Saharan Africa	–4.8	–5.3	–4.4	–6.1
South Asia	+13.8	+34.1	+36.8	+32.0
East Asia (excluding China and Japan)	+6.7	+23.0	+26.4	+20.1
Japan	+5.5	+22.2	+26.0	+18.9
China	+5.5	+22.3	+26.1	+19.1
Observed change, 1990 to 2000	+5.6	+23.8	+26.2	+19.4

NOTES: Estimates are based on life expectancy and population data for 169 countries from World Bank (2002); see Table 1 for list of countries. For each region, the change in inequality is recalculated under the assumption that life expectancy in that region had risen at the world-average rate. See text for details.

The larger the difference between observed and predicted values, the larger the contribution of the region to the change in overall health inequality. To illustrate, consider the case of life expectancy growth in China from 1990 to 2000. The predicted change (+5.5 percent as measured by the Gini coefficient) differs very little from the actual change (+5.6 percent as measured by the Gini coefficient). Clearly, then, changing life expectancy in China did not contribute appreciably to the reversal of the trend in between-country and between-region health inequality in the 1990s. The same is true for life expectancy in the West, Latin America, the Middle East and North Africa, Japan, and East Asia (outside China). What occurred in the 1990s to arrest the decline in between-country health inequality was of course the worsening of life expectancy in sub-Saharan Africa. Had life expectancy in sub-Saharan Africa during this period increased by 1.9 percent (the average for the rest of the world), between-country health inequality would have continued to decline—by 4.8 percent as measured by the Gini coefficient, by 5.3 percent by Theil, by 4.4 percent by MLD, or by 6.1 percent by CV^2.

The increased population share of sub-Saharan Africa also contributed to the rise in between-country inequality. Despite declining life expectancy, the population of sub-Saharan Africa continued to grow faster than the world average because of the region's high fertility rates. Table 4 quantifies the resulting allocation effect. Faster-than-world-average population growth in

TABLE 4 Allocation effects on health inequality, by region, 1990–2000: Predicted change in indexes of inequality with rate of growth of population set at the world average (percent)

	Index			
Region	Gini coefficient	Theil index	MLD	CV^2
Western Europe	+5.7	+22.7	+26.5	+19.5
Transition economies	+5.0	+21.3	+25.1	+18.1
Western offshoots	+5.7	+22.5	+26.3	+19.3
Latin America and the Caribbean	+5.8	+22.6	+26.4	+19.4
Middle East and North Africa	+5.8	+22.7	+26.5	+19.5
Sub-Saharan Africa	+2.0	+15.5	+19.0	+12.5
South Asia	+5.6	+22.4	+26.3	+19.2
East Asia (excluding China and Japan)	+5.8	+22.6	+26.4	+19.4
Japan	+5.8	+22.8	+26.5	+19.6
China	+5.3	+21.6	+25.4	+18.4
Observed change, 1990 to 2000	+5.6	+23.8	+26.2	+19.4

NOTES: Estimates are based on life expectancy and population data for 169 countries from World Bank (2002); see Table 1 for list of countries. For each region, the change in inequality is recalculated under the assumption that population in that region had grown at the world-average rate. See text for details.

sub-Saharan Africa boosted inequality by fattening the lower tail of the cross-national life expectancy distribution. Because life expectancy is so much lower in sub-Saharan Africa than it is in the rest of the world—in 2000, more than 15 years below the next-lowest region, South Asia—health inequality across countries increases when populations grow faster in Africa than in other regions.

In short, sub-Saharan Africa is central to the story of between-country health inequality in the late twentieth century. Even though only about one-tenth of the world's people live in sub-Saharan Africa, trends in this region in the 1990s were responsible for reversing a half-century trend toward equality in life expectancy across countries. Because life expectancy in sub-Saharan Africa is by far the lowest in the world, inequality in life expectancy across countries is boosted by that region's slower-than-world-average growth in life expectancy and faster-than-world-average growth in population. Both occurred during the 1990s.

The inequality-boosting effect of adverse health trends in sub-Saharan Africa was partly offset by positive trends in South Asia. In 1980, life expectancies in sub-Saharan Africa and in South Asia were about six years apart: 47.6 years in Africa, 53.4 years in South Asia (Table 1). Since 1980 the gap has more than doubled, with life expectancy in South Asia increasing to 62.1 years by 2000 and in sub-Saharan Africa falling to 46.5 years. Because South Asia's population is so large and because its life expectancy

TABLE 5 Leading contributors to change in between-country health inequality, 1990–2000

	Effect on between-country inequality
1. Sub-Saharan Africa, declining life expectancy	Increased inequality
2. South Asia, rising life expectancy	Reduced inequality
3. Sub-Saharan Africa, faster-than-world-average population growth	Increased inequality

SOURCE: Tables 3 and 4.

is moving up toward the world average, the life expectancy trend in South Asia has a palpable growth effect on the between-country inequality trend. In this case, however, the effect is to reduce inequality. Had life expectancy in South Asia not grown faster than the world average, inequality would have risen faster than it did in the 1990s: the Gini coefficient, for example, would have increased by 13.8 percent rather than the actual rise of 5.6 percent (Table 3).

It can be seen from Tables 3 and 4 that the growth effects for specific regions are generally larger than the allocation effects, consistent with our earlier finding that a growth effect—not an allocation effect—accounts for most of the rise in between-country inequality. Inequality increased in the 1990s primarily because life expectancies grew at different rates across countries, and only secondarily because populations grew at different rates.

Table 5 summarizes these results by listing the three most consequential factors in the recent inequality trend. The inequality-depressing effects of rising life expectancy in South Asia were more than offset by the inequality-boosting effects of declining life expectancy and faster-than-world-average population growth in sub-Saharan Africa. As a result, inequality increased overall.

Future health inequality

Average life expectancy is now above the world average in all major regions except South Asia and sub-Saharan Africa. Since life expectancy is growing rapidly in South Asia, sub-Saharan Africa will likely soon be the only region in the bottom half of the international distribution of life expectancy. Given such a highly skewed distribution, the sources for future declines in between-country health inequality will be limited to (1) faster-than-world-average gains in life expectancy in sub-Saharan Africa, (2) sharp drops in life expectancy in other world regions, or (3) a sharp drop in Africa's population share. Barring any major social or economic catastrophe, the second source is unlikely, so sub-Saharan Africa is the key to the future between-country inequality trend. Only if the trend of declining life expectancy in sub-Saharan Africa is reversed or its population share is greatly reduced will between-country inequality be likely to track downward again.

The 2002 projections of the United Nations Population Division (UNPD) (2003) assume that life expectancy at birth in Africa will continue to decline over the next five or ten years, then start climbing again after about 2010—reaching 65 years by the middle of the twenty-first century, about 18 years greater than its present level. If these projections hold, we can expect the trend of rising between-country health inequality to persist for at least five or ten more years. After that point, inequality will track downward again, just as it did through much of the past half-century. It is important to note, however, that these projections assume that HIV/AIDS prevalence rates in Africa will peak some time before 2010, then decline over the next four decades (UNPD 2003: Table 17). The current trend of rising between-country inequality is likely to persist for a longer period if the assumed declines in HIV/AIDS prevalence rates do not come about so soon.

Finally, what do these results mean for the trend in global health inequality? Recall (from equation 1) that global health inequality is the sum of population-weighted between-country and within-country inequality. If between-country health inequality is much larger than within-country inequality or if within-country inequality has changed only trivially since 1990, then global health inequality is rising. But if within-country inequality is the larger component of global health inequality, and if within-country inequality has declined, then the trend in global health inequality could be downward despite the rise in between-country inequality. In a recent study of global health inequality, Pradhan, Sahn, and Younger (2003) argue that within-country inequality now accounts for the bulk of total global health inequality. If within-country inequality in fact is declining, it is possible that global health inequality is declining even though between-country inequality is rising. The results of Pradhan and coauthors, however, are based on analyses of the global distribution of children's height—a different measure of health. It may be inappropriate to combine their results with ours to infer the direction of the global trend. A confident assessment of the recent trend in global health inequality must await further research.

Notes

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1 An inverted-U-shaped trend in health inequality is also reported by Bourguignon and Morrisson (2002: Figure 3), in their recent study of inequality in global living standards over the nineteenth and twentieth centuries. According to their estimates, the turning point occurred around 1930. Between-country inequality was no greater in the late twentieth century than it was in the early nineteenth.

2 Throughout this note, life expectancy data are given for both sexes combined, but

the results are similar when divided by sex. China and Japan are separately identified: China because of its size, Japan because its life expectancy has historically been higher than in most other East Asian countries.

3 Good introductions to the large technical literature on measuring inequality include Allison (1978), Cowell (1995), Jenkins (1991), Schwartz and Winship (1980), and Sen (1997). For the application of inequality indexes to health-related data, see Gakidou et al. (2000), Gakidou and King (2002), Murray et al. (1999), Shkolnikov et al. (2003), and Wilmoth and Horiuchi (1999).

4 Technically, the Gini coefficient is the most sensitive of the four indexes to the mode of the distribution (Allison 1978: 868); but for the distribution of life expectancy across countries, the mode is closer to the middle of the distribution than to either extreme tail, making it appropriate to associate the Gini coefficient with the middle of the distribution.

5 It is important to weight countries by population in this study because a goal is to analyze the trend in global health inequality, and global inequality is the sum of *population-weighted* between-country and within-country inequality, as described above (equation 1). Sensitivity analyses (not shown, but available from the authors) find that the main results of the study do not change when we weight countries equally.

6 The decomposition of the MLD given in both Mookherjee and Shorrocks (1982) and Jenkins (1995) has four additive terms—two for between-group inequality and two for within-group inequality. For the present analysis, the two within-group inequality terms drop out, since our focus is on inequality between—not within—countries. The two terms given in equation 4 here correspond to Mookherjee and Shorrocks's equations 14c and 14d and to "term C" and "term D" in Jenkins's equation 5.

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